

THIN-FILM DEPOSITION CONTROLLER Using Microcontroller

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This project is used to control the movement of the spray head and nozzle in a thin metal or insulation film deposition system. Here, AT89C51 or P89V51RD2 microcontroller (MCU) is used to control the spray head and nozzle in x and y directions. It is an inexpensive, simple and efficient solution for a flat surface. The setup of the thin-film deposition controller using MCU is shown in Fig. 1.

In this technique, the spray head—which contains fine sprays of molten metal for spray coating over a flat surface—moves in x and y directions on the sheet of a given job. It is controlled and guided by an MCU using two motors and four limit switches with mechanical support structure, as shown in Fig. 1. Block diagram of the thin-film deposition controller using MCU is shown in Fig. 2. Flow chart of the circuit is shown in Fig. 3.

Circuit and working

The circuit diagram of the thin-film deposition technique using MCU is shown in Fig. 4. It is built around AT89C51 MCU (IC1), 16x2 LCD (LCD1), six 5V, 1CO relays (RL1 through RL6), six 1N4007 rectifier diodes (D1-D6), six LEDs (LED1 through LED6), six pnp transistors (T1 through T6), two low-rpm 3-phase motors along with 3-phase contactors, DC-operated nozzle, limit switches (S1 through S4) and a few other components.

The spray head is moved in x and y directions using two 3-phase motors (M1 and M2), controlled by the MCU in clockwise (CW) and counter-clockwise (CCW) directions, as shown in Fig. 1. The motors have shafts with fine threads.

Here, AT89C51 is the main controller of the circuit. It controls 5V relay units from its port pins P1.0 through

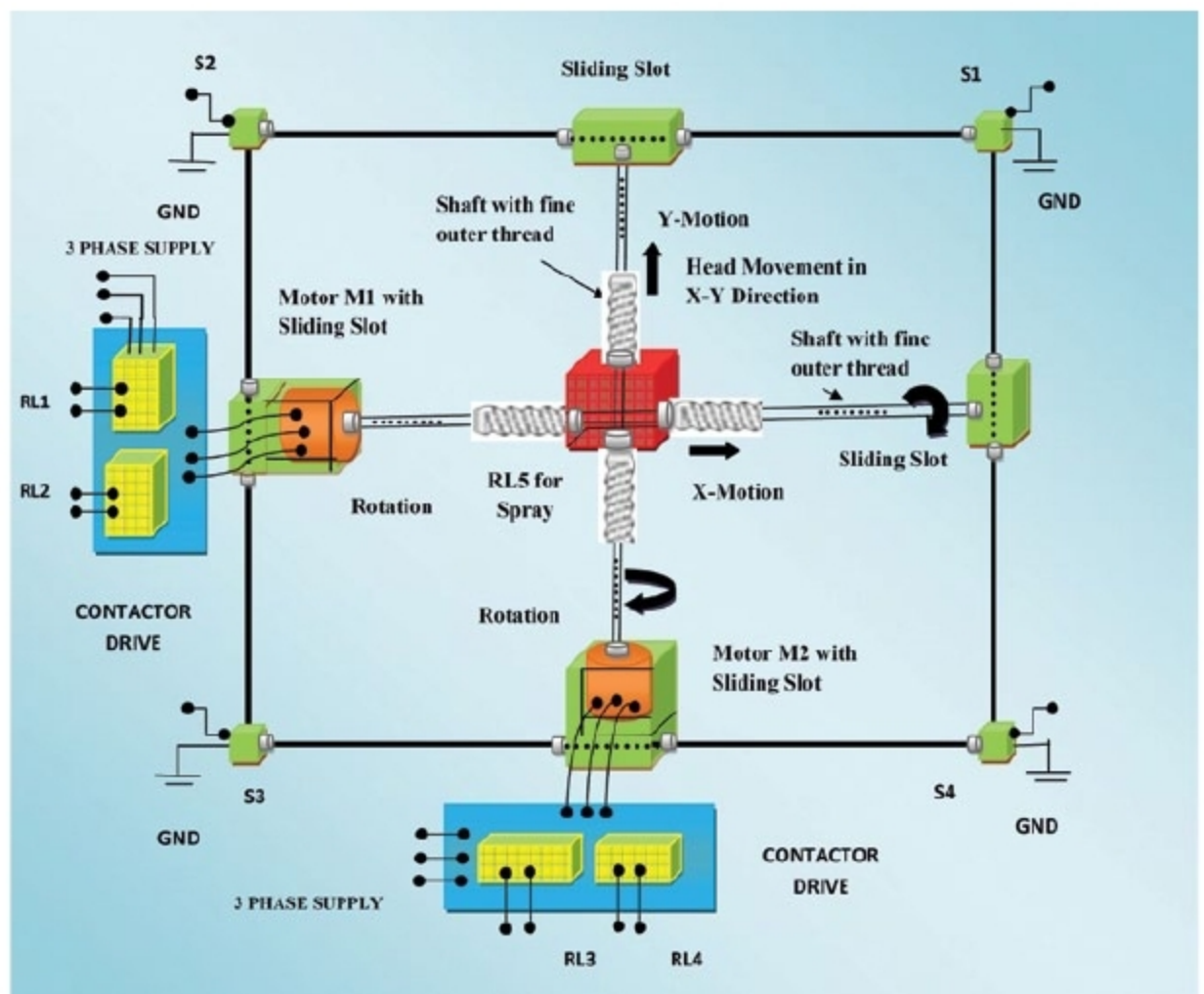


Fig. 1: Setup of thin-film deposition controller using MCU

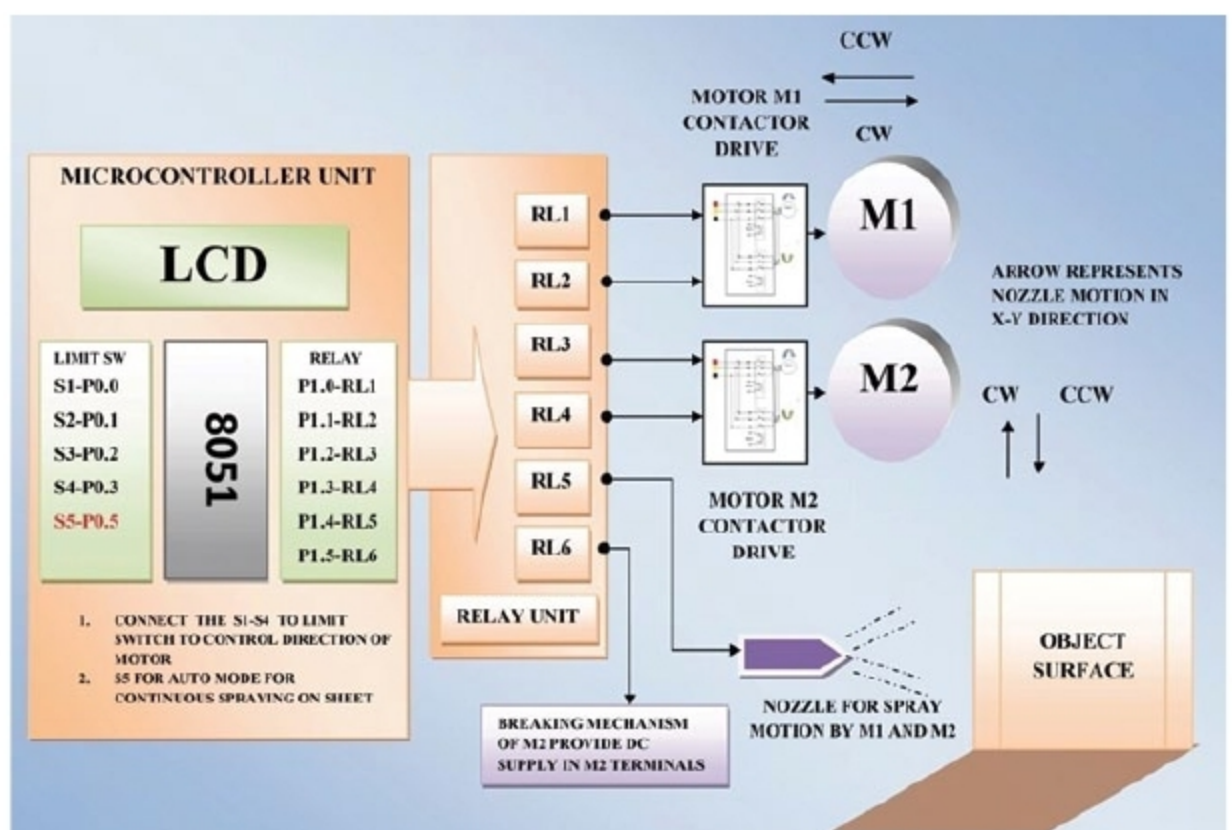


Fig. 2: Block diagram of thin-film deposition controller using MCU